

SUMMARY

Postdoctoral Research Fellow specialising in trustworthy and reliable machine learning and large language model safety, with a track record of designing novel algorithms and advancing AI security research through adversarial analysis of model behaviour. Experienced in multidisciplinary, collaborative research spanning academia, government, and industry partners, alongside current teaching and research-project coordination in Applied AI/ML. Seeking to build an independent, internationally recognised research program while contributing to high-quality, blended teaching and Honours/HDR supervision.

EDUCATION

- 2019 - 2023 Ph.D., Computer Science, Adelaide University, Australia**
Topic: Trustworthy and Reliable Machine Learning (AI Safety)
- 2013 - 2014 Master of Electronic and Computer Engineering, RMIT Vietnam University, Vietnam**
Major: Computer Engineering
- 2007 - 2012 Bachelor of Electrical and Electronic Engineering, Ho Chi Minh City University of Technology**
Major: Electrical and Electronic Engineering

RESEARCH EXPERIENCE

- 2023 – Present Postdoc Fellow, Adelaide University & Australian Institute for Machine Learning**
LLM Safety & Security (Collaborate with Monash University):
 - Investigated vulnerabilities of LLMs under jailbreak attacks and developed detection methods achieving up to 95% accuracy through systematic adversarial evaluation of LLMs, contributing to AI safety, robustness, and responsible deployment.**Certified Defense Robustness (Collaborate with DSTG):**
 - Developed a novel adversarial attack to systematically investigate vulnerabilities and fundamental limitations in robustness guarantees of certified defense mechanisms.**Efficient Domain Adaptation (Collaborate with DSTG):**
 - Developed a novel few-shot learning and parameter-efficient algorithm for vision-language foundation models to achieve 20x fine-tuning efficiency gains without performance drop, even with limited labelled data.**LLM Agents for Surveillance Systems (Collaborate with Industry):**
 - Designed and implemented an LLM agent framework for automated data extraction, information integration and structured output generation, supporting real-time decision-making in naval surveillance systems.
- 2019 – 2023 PhD Candidate, Adelaide University**
Black-Box Robustness of Deep Neural Networks: Investigated the security of deep learning architectures under Black-Box threat models, simulating real-world deployment constraints.
 - Designed multiple query-efficiency attack paradigms (RamBoAttack, SparseEvoAttack and BruSLeAttack) based on Bayesian optimization or Evolutionary algorithms with low perturbation. These attacks can achieve 10× fewer queries than state-of-the-art methods and 5% higher attack success rates in large-scale search spaces.
 - Developed a defense mechanism based on diversity and randomness to enhance the robustness of models against black-box attacks.

TEACHING EXPERIENCE

- 2026 Lecturer:
- Applied Artificial Intelligence and Machine Learning (COMP 6026)
 - Artificial Intelligence and Machine Learning Research Project Part A&B (COMP 7003, COMP 7000): Supervised students in Responsible AI & LLM Safety; Security Vulnerabilities in Autonomous LLM Agents.

TEACHING EXPERIENCE

- 2023 Teaching Assistant:
- Introduction to Statistical Machine Learning (COMP SCI 3314/7314)
 - Operating Systems (COMP SCI 3004/7064)
 - Grand Challenges in Computer Science (COMP SCI 1104)

SELECTED PUBLICATIONS

- 2026 **Quoc Viet V.**, Ehsan A., and Damith R. "Can Model Randomization Offer Robustness Against Query-Based Black-Box Attacks?", IEEE Transactions on Information Forensics and Security (TIFS) (Under Preparation for Submission) (CORE A*).
- 2026 **Quoc Viet V.**, Trung Le, Damith R. and , Ehsan A. "SAFEGuard: Detect Optimization-Based Jailbreak Attacks Through Harmful Semantic Analysis and Fluency Measurement", Empirical Methods in Natural Language Processing (EMNLP 2026) (Under Review) (CORE A*).
- 2026 **Quoc Viet V.**, Tashreque M., Paul M., Tamas A., Ehsan A., and Damith R. "Exploring the Manipulation of Certified Model Robustness Guarantees", IEEE Transactions On Pattern Analysis And Machine Intelligence (TPAMI) (Under Preparation for Submission) (CORE A*).
- 2026 **Quoc Viet V.**, Tashreque M., Paul M., Tamas A., Ehsan A., and Damith R. "Certified but Fooled! Breaking Certified Defences with Ghost Certificates", The Fortieth AAAI Conference on Artificial Intelligence (AAAI) (CORE A*).
- 2024 **Quoc Viet V.**, Ehsan A., and Damith R. "BruSLeAttack: Query-Efficient Score-Based Sparse Adversarial Attack", International Conference on Learning Recognition (ICLR) (CORE A*).
- 2022 **Quoc Viet V.**, Ehsan A., and Damith R. "Query efficient decision based sparse attacks against black-box deep learning models", International Conference on Learning Recognition (ICLR) (CORE A*).
- 2022 **Quoc Viet V.**, Ehsan A., and Damith R. "RamBoAttack: A Robust Query Efficient Deep Neural Network Decision Exploit", Network and Distributed Systems Security Symposium (NDSS) (CORE A*).

AWARDS

- 2022 **NDSS Student travel Grant** is provided by Network and Distributed System Security Symposium for emerging security researchers who could contribute to the published content at NDSS.
- 2019 **Postgraduate Research Scholarship** is provided by the Faculty of Engineering, Computer & Mathematical Sciences Divisional for talented students worldwide who want to carry out cutting-edge research and make a real impact on the world.
- 2012 **Intel Scholarship** is provided by Intel Vietnam for the brightest and best students who will be the core technical resource for Intel Vietnam and the high-tech industry of Vietnam.

PROFESSIONAL SERVICES, INDUSTRY ENGAGEMENT

- Peer reviewer for leading international venues ICLR, TNNLS, NeurIPS, ACL, ICML.
- Collaborative research partnerships with the Defence Science and Technology Group (DSTG) and industry.

SKILLS AND INTERESTS

• **Machine learning and Data:** PyTorch, Numpy, Pandas, Scikit-Learn, Jupyter Lab, Hugging Face.

• **Interests:** Trustworthy and Responsible AI, Large Language Models, Agentic AI.